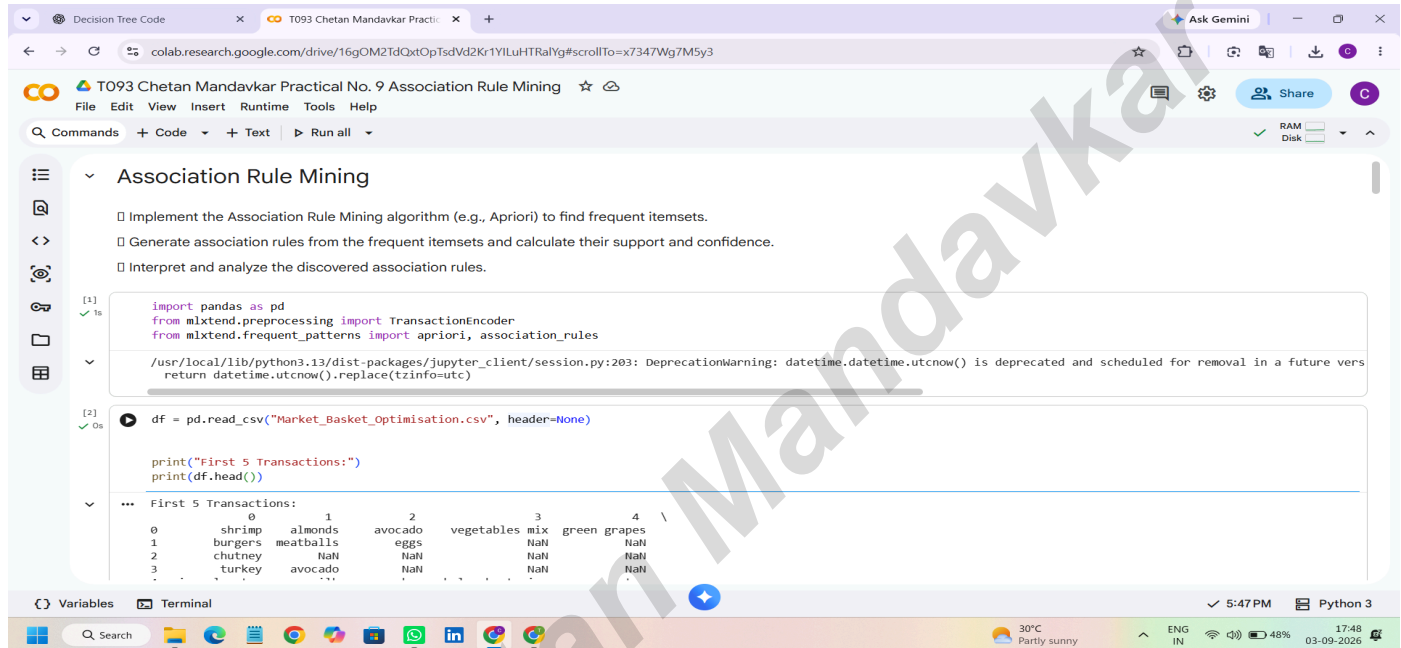


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Aim:- Association Rule Mining

Implement the Association Rule Mining algorithm (e.g., Apriori) to find frequent itemsets. Generate association rules from the frequent itemsets and calculate their support and confidence. Interpret and analyze the discovered association rules.



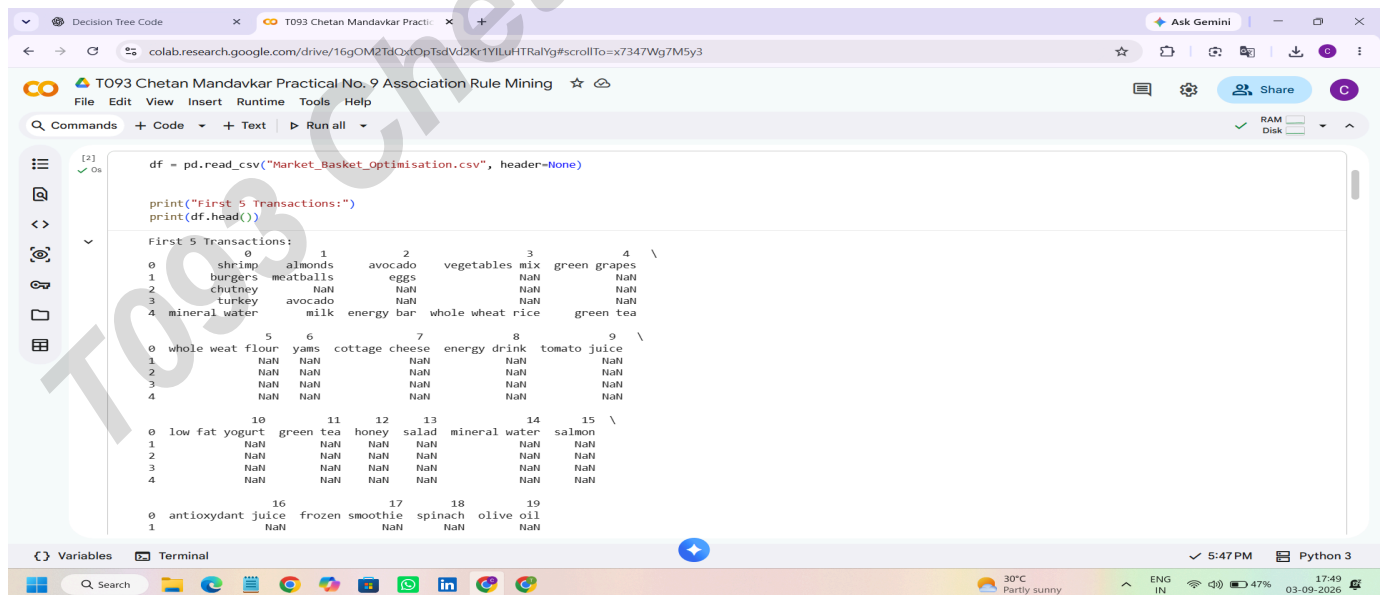
```
[1] In
import pandas as pd
from mlxtend.preprocessing import TransactionEncoder
from mlxtend.frequent_patterns import apriori, association_rules

/usr/local/lib/python3.13/dist-packages/jupyter_client/session.py:203: DeprecationWarning: datetime.datetime.utcnow() is deprecated and scheduled for removal in a future version. Use datetime.datetime.now(datetime.timezone.utc) instead.
return datetime.datetime.utcnow().replace(tzinfo=utc)

[2] Out
df = pd.read_csv("Market_Basket_Optimisation.csv", header=None)

print("First 5 Transactions:")
print(df.head())
```

	0	1	2	3	4
0	shrimp	almonds	avocado	vegetables mix	green grapes
1	burgers	meatballs	eggs	NaN	NaN
2	chutney	NaN	NaN	NaN	NaN
3	turkey	avocado	NaN	NaN	NaN
4	NaN	NaN	NaN	NaN	NaN



```
[2] Out
df = pd.read_csv("Market_Basket_Optimisation.csv", header=None)

print("First 5 Transactions:")
print(df.head())
```

	0	1	2	3	4
0	shrimp	almonds	avocado	vegetables mix	green grapes
1	burgers	meatballs	eggs	NaN	NaN
2	chutney	NaN	NaN	NaN	NaN
3	turkey	avocado	NaN	NaN	NaN
4	mineral water	milk	energy bar	whole wheat rice	green tea
5	whole wheat flour	yams	cottage cheese	energy drink	tomato juice
6	NaN	NaN	NaN	NaN	NaN
7	NaN	NaN	NaN	NaN	NaN
8	NaN	NaN	NaN	NaN	NaN
9	NaN	NaN	NaN	NaN	NaN
10	low fat yogurt	green tea	honey	salad	mineral water
11	NaN	NaN	NaN	NaN	NaN
12	NaN	NaN	NaN	NaN	NaN
13	NaN	NaN	NaN	NaN	NaN
14	NaN	NaN	NaN	NaN	NaN
15	NaN	NaN	NaN	NaN	NaN
16	antioxydant juice	frozen smoothie	spinach	olive oil	NaN
17	NaN	NaN	NaN	NaN	NaN
18	NaN	NaN	NaN	NaN	NaN
19	NaN	NaN	NaN	NaN	NaN

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Roll No. T093

Subject:- Artificial Intelligence

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T093 Chetan Mandavkar Practical No. 9 Association Rule Mining

File Edit View Insert Runtime Tools Help

Commands + Code + Text Run all

RAM Disk

```
... 0 whole wheat flour 5 6 cottage cheese 7 energy drink 8 tomato juice 9 \
1 NaN NaN NaN NaN NaN
2 NaN NaN NaN NaN NaN
3 NaN NaN NaN NaN NaN
4 NaN NaN NaN NaN NaN

0 low fat yogurt 10 11 green tea 12 honey 13 salad 14 mineral water 15 salmon \
1 NaN NaN NaN NaN NaN NaN
2 NaN NaN NaN NaN NaN NaN
3 NaN NaN NaN NaN NaN NaN
4 NaN NaN NaN NaN NaN NaN

0 antioxydant juice 16 17 frozen smoothie 18 spinach 19 olive oil
1 NaN NaN NaN NaN
2 NaN NaN NaN NaN
3 NaN NaN NaN NaN
4 NaN NaN NaN NaN

/usr/local/lib/python3.13/dist-packages/jupyter_client/session.py:203: DeprecationWarning: datetime.datetime.utcnow() is deprecated and scheduled for removal in a future version. Use datetime.datetime.now(datetime.UTC) instead.
return datetime.datetime.utcnow().replace(tzinfo=utc)
/usr/local/lib/python3.13/dist-packages/jupyter_client/session.py:203: DeprecationWarning: datetime.datetime.utcnow() is deprecated and scheduled for removal in a future version. Use datetime.datetime.now(datetime.UTC) instead.
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/usr/local/lib/python3.13/dist-packages/jupyter_client/session.py:203: DeprecationWarning: datetime.datetime.utcnow() is deprecated and scheduled for removal in a future version. Use datetime.datetime.now(datetime.UTC) instead.
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return datetime.datetime.utcnow().replace(tzinfo=utc)
/usr/local/lib/python3.13/dist-packages/jupyter_client/session.py:203: DeprecationWarning: datetime.datetime.utcnow() is deprecated and scheduled for removal in a future version. Use datetime.datetime.now(datetime.UTC) instead.
return datetime.datetime.utcnow().replace(tzinfo=utc)
/usr/local/lib/python3.13/dist-packages/jupyter_client/session.py:203: DeprecationWarning: datetime.datetime.utcnow() is deprecated and scheduled for removal in a future version. Use datetime.datetime.now(datetime.UTC) instead.
return datetime.datetime.utcnow().replace(tzinfo=utc)
```

Variables Terminal

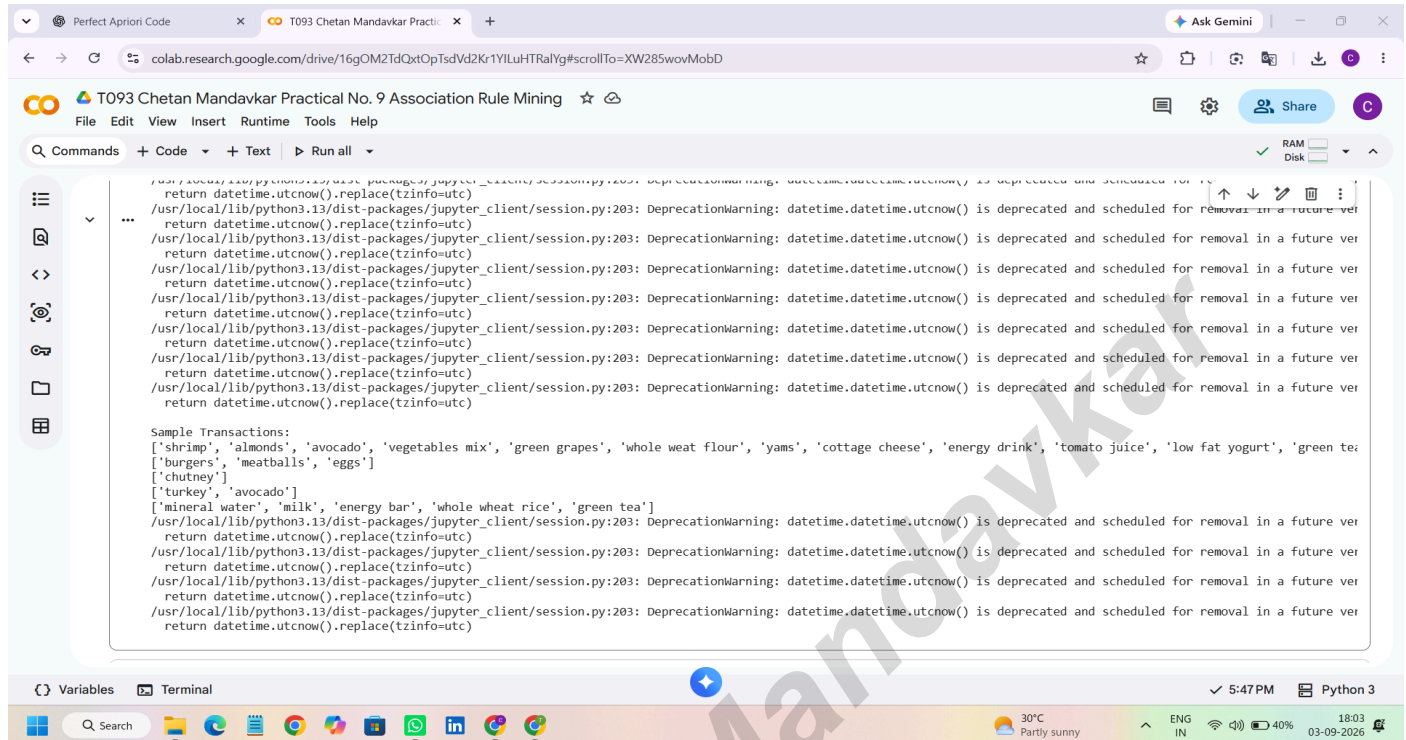
5:47 PM Python 3

30°C Partly sunny ENG IN 47% 17:50 03-09-2026

[illegible]

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Roll No. T093
Subject:- Artificial Intelligence

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```
from sklearn.preprocessing import TransactionEncoder

te = TransactionEncoder()
te_array = te.fit(transactions).transform(transactions)

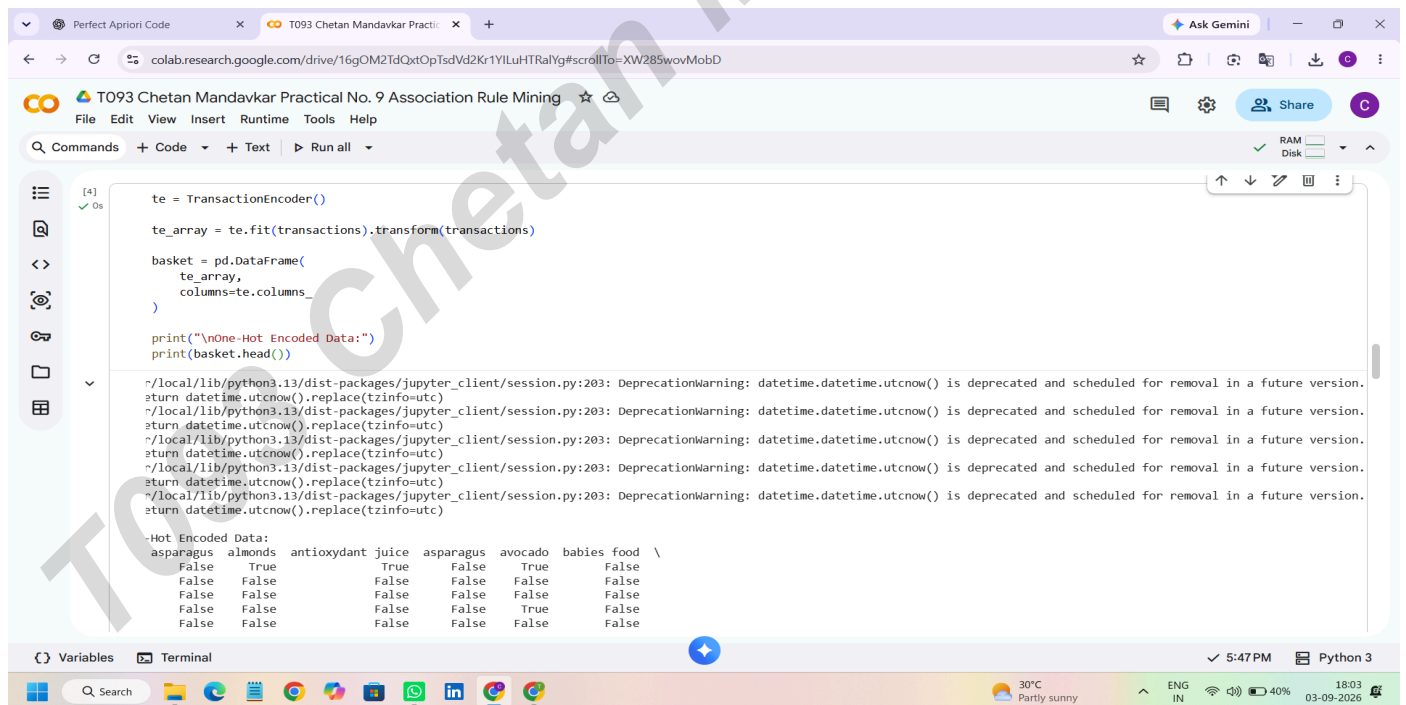
basket = pd.DataFrame(
    te_array,
    columns=te.columns_
)

print("\nOne-Hot Encoded Data:")
print(basket.head())
```

Sample Transactions:

```
['shrimp', 'almonds', 'avocado', 'vegetables mix', 'green grapes', 'whole weat flour', 'yams', 'cottage cheese', 'energy drink', 'tomato juice', 'low fat yogurt', 'green tea']
['burgers', 'meatballs', 'eggs']
['chutney']
['turkey', 'avocado']
['mineral water', 'milk', 'energy bar', 'whole wheat rice', 'green tea']
```

	shrimp	almonds	avocado	vegetables mix	green grapes	whole weat flour	yams	cottage cheese	energy drink	tomato juice	low fat yogurt	green tea
0	True	True	True	True	True	True	True	True	True	True	True	True
1	True	True	True	True	True	True	True	True	True	True	True	True
2	True	True	True	True	True	True	True	True	True	True	True	True
3	True	True	True	True	True	True	True	True	True	True	True	True
4	True	True	True	True	True	True	True	True	True	True	True	True



```
te = TransactionEncoder()
te_array = te.fit(transactions).transform(transactions)

basket = pd.DataFrame(
    te_array,
    columns=te.columns_
)

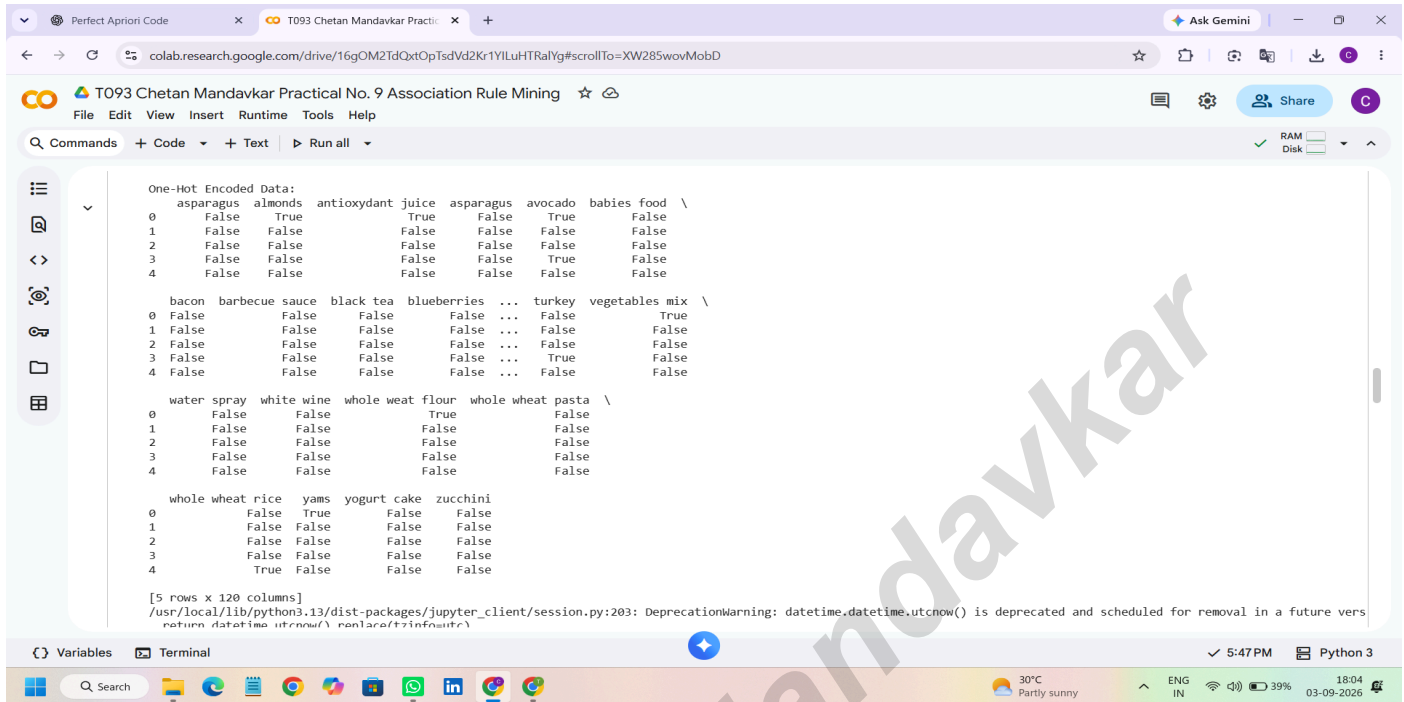
print("\nOne-Hot Encoded Data:")
print(basket.head())
```

One-Hot Encoded Data:

	shrimp	almonds	avocado	vegetables mix	green grapes	whole weat flour	yams	cottage cheese	energy drink	tomato juice	low fat yogurt	green tea
0	True	True	True	True	True	True	True	True	True	True	True	True
1	True	True	True	True	True	True	True	True	True	True	True	True
2	True	True	True	True	True	True	True	True	True	True	True	True
3	True	True	True	True	True	True	True	True	True	True	True	True
4	True	True	True	True	True	True	True	True	True	True	True	True

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The screenshot shows a Jupyter Notebook titled "T093 Chetan Mandavkar Practical No. 9 Association Rule Mining". The notebook contains a dataset with 5 rows and 120 columns. The data is displayed in a table format with columns grouped by product categories. The categories and their items are: asparagus, almonds, antioxidant, juice, asparagus, avocado, babies food; bacon, barbecue, sauce, black tea, blueberries, turkey, vegetables mix; water, spray, white wine, whole wheat, flour, whole wheat, pasta; and whole wheat, rice, yams, yogurt, cake, zucchini. The data is binary, with values being either True or False. The notebook interface includes a menu bar (File, Edit, View, Insert, Runtime, Tools, Help), a toolbar (Commands, Code, Text, Run all), and a status bar (Variables, Terminal, 5:47 PM, Python 3).

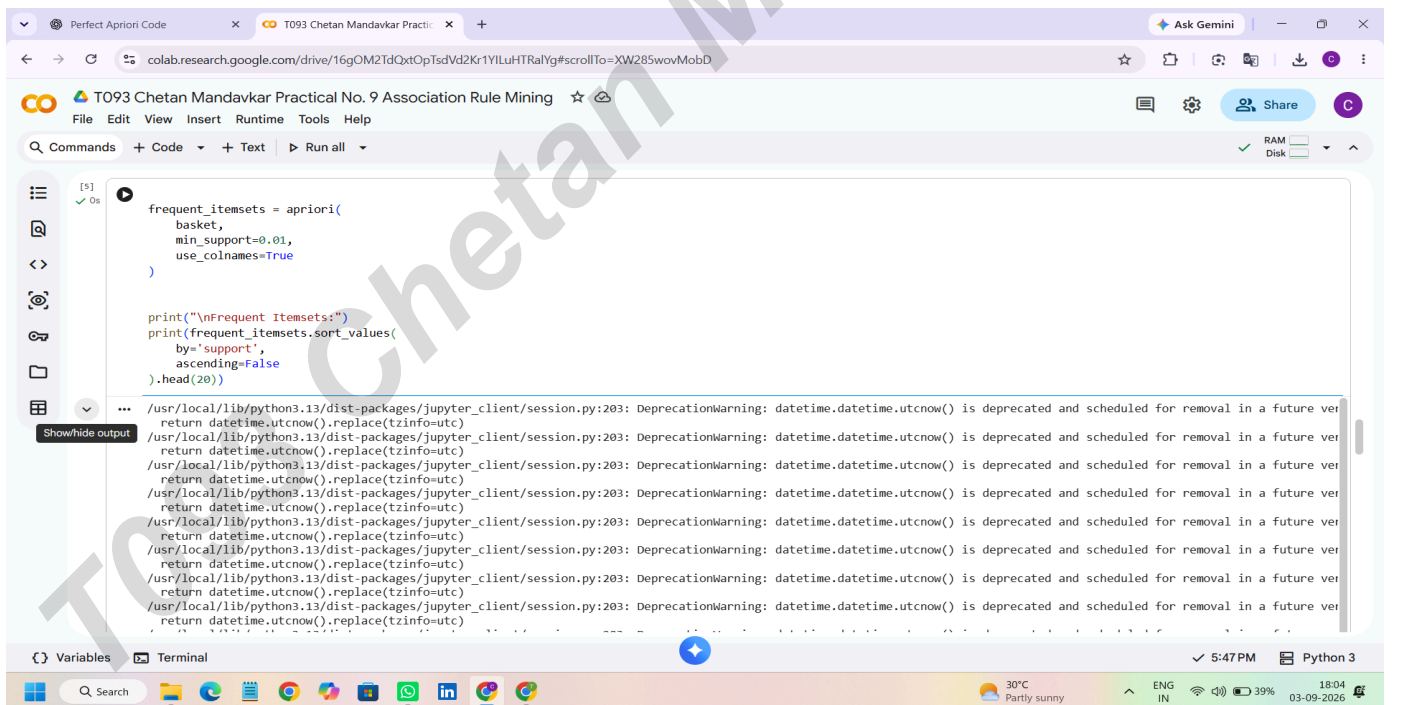
```
One-Hot Encoded Data:
asparagus  almonds  antioxidant  juice  asparagus  avocado  babies food \
0      False      True      False      True      False      True      False
1      False      False      False      False      False      False      False
2      False      False      False      False      False      False      False
3      False      False      False      False      False      True      False
4      False      False      False      False      False      False      False

bacon  barbecue  sauce  black tea  blueberries  ...  turkey  vegetables mix \
0      False      False      False      False      ...      False      True
1      False      False      False      False      ...      False      False
2      False      False      False      False      ...      False      False
3      False      False      False      False      ...      True      False
4      False      False      False      False      ...      False      False

water  spray  white wine  whole wheat  flour  whole wheat  pasta \
0      False      False      False      True      False      False
1      False      False      False      False      False      False
2      False      False      False      False      False      False
3      False      False      False      False      False      False
4      False      False      False      False      False      False

whole wheat  rice  yams  yogurt  cake  zucchini
0      False      True      False      False      False      False
1      False      False      False      False      False      False
2      False      False      False      False      False      False
3      False      False      False      False      False      False
4      True      False      False      False      False      False

[5 rows x 120 columns]
/usr/local/lib/python3.13/dist-packages/jupyter_client/session.py:203: DeprecationWarning: datetime.datetime.utcnow() is deprecated and scheduled for removal in a future version. Use datetime.datetime.now(datetime.timezone.utc) instead.
```



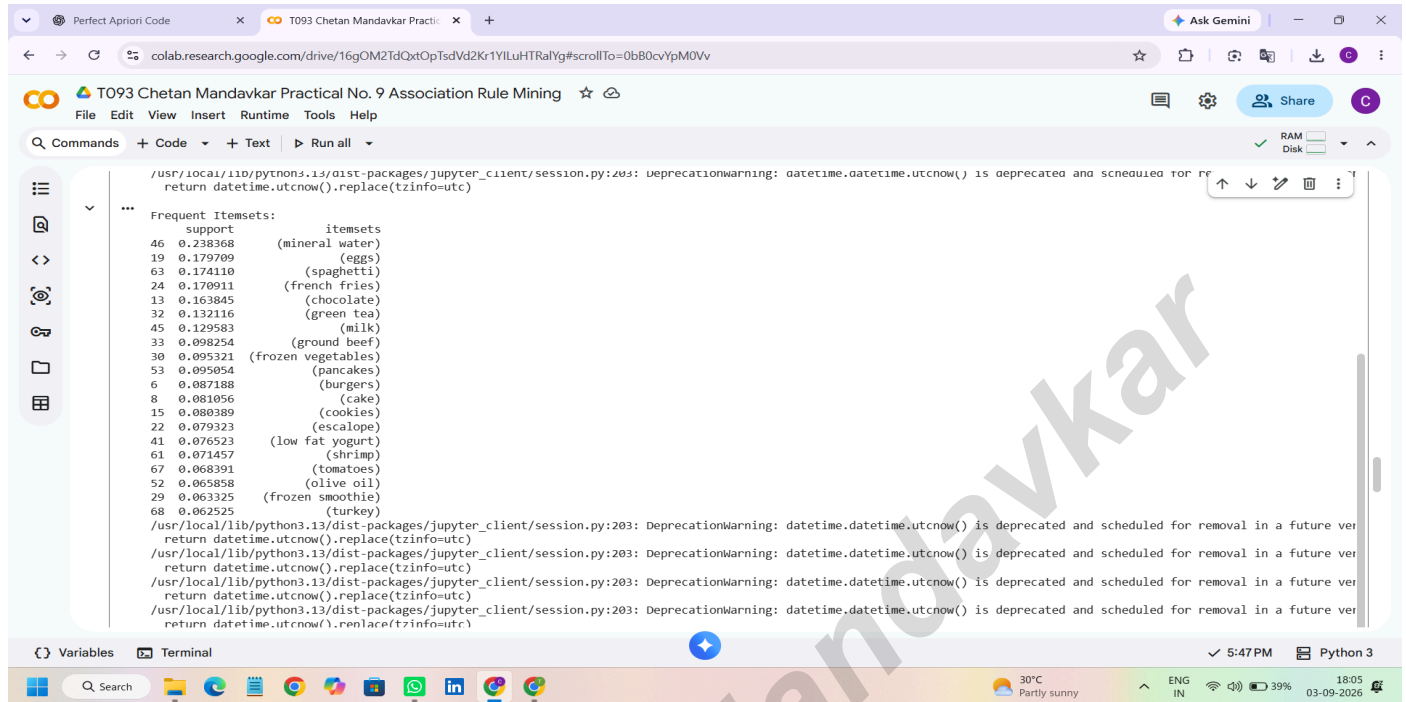
The screenshot shows the same Jupyter Notebook with the execution of the Apriori algorithm. The code defines the frequent itemsets using the Apriori function from the mlxtend library. The parameters are: basket, min_support=0.01, and use_colnames=True. The output is printed, showing the frequent itemsets sorted by support. The notebook interface is the same as the previous screenshot.

```
frequent_itemsets = apriori(
    basket,
    min_support=0.01,
    use_colnames=True
)

print("\nFrequent Itemsets:")
print(frequent_itemsets.sort_values(
    by='support',
    ascending=False
).head(20))
```

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Roll No. T093
Subject:- Artificial Intelligence

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The screenshot shows a Jupyter Notebook interface with a code cell containing the output of the `frequent_itemsets` function. The output is a list of itemsets with their support and confidence values. The itemsets are: (mineral water), (eggs), (spaghetti), (french fries), (chocolate), (green tea), (milk), (ground beef), (frozen vegetables), (pancakes), (burgers), (cake), (cookies), (escalope), (low fat yogurt), (shrimp), (tomatoes), (olive oil), (frozen smoothie), and (turkey). The support values range from 0.062525 to 0.238368, and the confidence values range from 0.065858 to 0.179709.

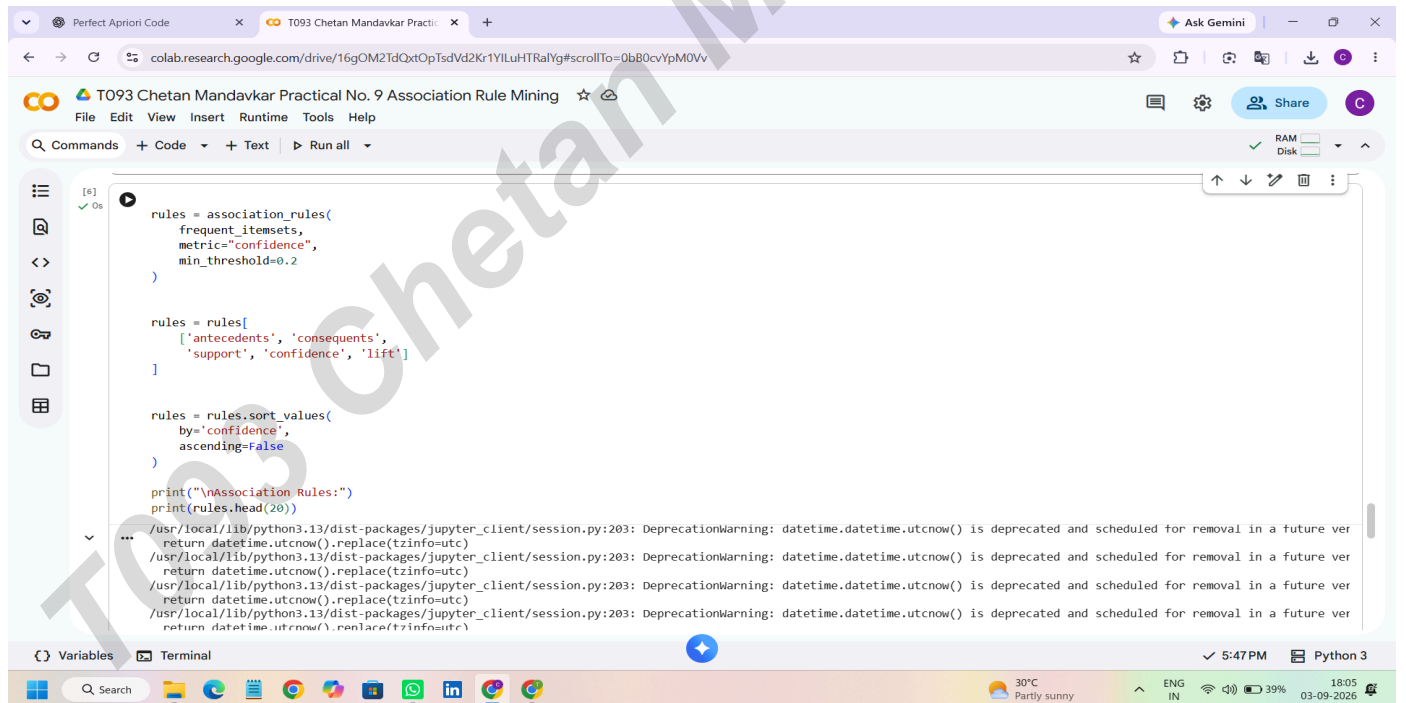
```
...
Frequent Itemsets:
support      itemsets
46 0.238368   (mineral water)
19 0.179709   (eggs)
63 0.174110   (spaghetti)
24 0.170911   (french fries)
13 0.163845   (chocolate)
32 0.132116   (green tea)
45 0.129583   (milk)
33 0.098254   (ground beef)
30 0.095321   (frozen vegetables)
53 0.095054   (pancakes)
6 0.087188    (burgers)
8 0.081056    (cake)
15 0.080389   (cookies)
22 0.079323   (escalope)
41 0.076523   (low fat yogurt)
61 0.071457   (shrimp)
67 0.068391   (tomatoes)
52 0.065858   (olive oil)
29 0.063325   (frozen smoothie)
68 0.062525   (turkey)

/usr/local/lib/python3.13/dist-packages/jupyter_client/session.py:203: DeprecationWarning: datetime.datetime.utcnow() is deprecated and scheduled for removal in a future version; use datetime.datetime.now(datetime.UTC) instead
  return datetime.datetime.utcnow().replace(tzinfo=utc)

/usr/local/lib/python3.13/dist-packages/jupyter_client/session.py:203: DeprecationWarning: datetime.datetime.utcnow() is deprecated and scheduled for removal in a future version; use datetime.datetime.now(datetime.UTC) instead
  return datetime.datetime.utcnow().replace(tzinfo=utc)

/usr/local/lib/python3.13/dist-packages/jupyter_client/session.py:203: DeprecationWarning: datetime.datetime.utcnow() is deprecated and scheduled for removal in a future version; use datetime.datetime.now(datetime.UTC) instead
  return datetime.datetime.utcnow().replace(tzinfo=utc)

/usr/local/lib/python3.13/dist-packages/jupyter_client/session.py:203: DeprecationWarning: datetime.datetime.utcnow() is deprecated and scheduled for removal in a future version; use datetime.datetime.now(datetime.UTC) instead
  return datetime.datetime.utcnow().replace(tzinfo=utc)
```



The screenshot shows a Jupyter Notebook interface with a code cell containing the output of the `association_rules` function. The output is a list of association rules with their antecedents, consequents, support, confidence, and lift values. The rules are: (mineral water) => (eggs), (mineral water) => (spaghetti), (mineral water) => (french fries), (mineral water) => (chocolate), (mineral water) => (green tea), (mineral water) => (milk), (mineral water) => (ground beef), (mineral water) => (frozen vegetables), (mineral water) => (pancakes), (mineral water) => (burgers), (mineral water) => (cake), (mineral water) => (cookies), (mineral water) => (escalope), (mineral water) => (low fat yogurt), (mineral water) => (shrimp), (mineral water) => (tomatoes), (mineral water) => (olive oil), (mineral water) => (frozen smoothie), (mineral water) => (turkey), (eggs) => (mineral water), (eggs) => (spaghetti), (eggs) => (french fries), (eggs) => (chocolate), (eggs) => (green tea), (eggs) => (milk), (eggs) => (ground beef), (eggs) => (frozen vegetables), (eggs) => (pancakes), (eggs) => (burgers), (eggs) => (cake), (eggs) => (cookies), (eggs) => (escalope), (eggs) => (low fat yogurt), (eggs) => (shrimp), (eggs) => (tomatoes), (eggs) => (olive oil), (eggs) => (frozen smoothie), (eggs) => (turkey), (spaghetti) => (mineral water), (spaghetti) => (eggs), (spaghetti) => (french fries), (spaghetti) => (chocolate), (spaghetti) => (green tea), (spaghetti) => (milk), (spaghetti) => (ground beef), (spaghetti) => (frozen vegetables), (spaghetti) => (pancakes), (spaghetti) => (burgers), (spaghetti) => (cake), (spaghetti) => (cookies), (spaghetti) => (escalope), (spaghetti) => (low fat yogurt), (spaghetti) => (shrimp), (spaghetti) => (tomatoes), (spaghetti) => (olive oil), (spaghetti) => (frozen smoothie), (spaghetti) => (turkey), (french fries) => (mineral water), (french fries) => (eggs), (french fries) => (spaghetti), (french fries) => (chocolate), (french fries) => (green tea), (french fries) => (milk), (french fries) => (ground beef), (french fries) => (frozen vegetables), (french fries) => (pancakes), (french fries) => (burgers), (french fries) => (cake), (french fries) => (cookies), (french fries) => (escalope), (french fries) => (low fat yogurt), (french fries) => (shrimp), (french fries) => (tomatoes), (french fries) => (olive oil), (french fries) => (frozen smoothie), (french fries) => (turkey), (chocolate) => (mineral water), (chocolate) => (eggs), (chocolate) => (spaghetti), (chocolate) => (french fries), (chocolate) => (green tea), (chocolate) => (milk), (chocolate) => (ground beef), (chocolate) => (frozen vegetables), (chocolate) => (pancakes), (chocolate) => (burgers), (chocolate) => (cake), (chocolate) => (cookies), (chocolate) => (escalope), (chocolate) => (low fat yogurt), (chocolate) => (shrimp), (chocolate) => (tomatoes), (chocolate) => (olive oil), (chocolate) => (frozen smoothie), (chocolate) => (turkey), (green tea) => (mineral water), (green tea) => (eggs), (green tea) => (spaghetti), (green tea) => (french fries), (green tea) => (chocolate), (green tea) => (milk), (green tea) => (ground beef), (green tea) => (frozen vegetables), (green tea) => (pancakes), (green tea) => (burgers), (green tea) => (cake), (green tea) => (cookies), (green tea) => (escalope), (green tea) => (low fat yogurt), (green tea) => (shrimp), (green tea) => (tomatoes), (green tea) => (olive oil), (green tea) => (frozen smoothie), (green tea) => (turkey), (milk) => (mineral water), (milk) => (eggs), (milk) => (spaghetti), (milk) => (french fries), (milk) => (chocolate), (milk) => (green tea), (milk) => (ground beef), (milk) => (frozen vegetables), (milk) => (pancakes), (milk) => (burgers), (milk) => (cake), (milk) => (cookies), (milk) => (escalope), (milk) => (low fat yogurt), (milk) => (shrimp), (milk) => (tomatoes), (milk) => (olive oil), (milk) => (frozen smoothie), (milk) => (turkey), (ground beef) => (mineral water), (ground beef) => (eggs), (ground beef) => (spaghetti), (ground beef) => (french fries), (ground beef) => (chocolate), (ground beef) => (green tea), (ground beef) => (milk), (ground beef) => (frozen vegetables), (ground beef) => (pancakes), (ground beef) => (burgers), (ground beef) => (cake), (ground beef) => (cookies), (ground beef) => (escalope), (ground beef) => (low fat yogurt), (ground beef) => (shrimp), (ground beef) => (tomatoes), (ground beef) => (olive oil), (ground beef) => (frozen smoothie), (ground beef) => (turkey), (frozen vegetables) => (mineral water), (frozen vegetables) => (eggs), (frozen vegetables) => (spaghetti), (frozen vegetables) => (french fries), (frozen vegetables) => (chocolate), (frozen vegetables) => (green tea), (frozen vegetables) => (milk), (frozen vegetables) => (ground beef), (frozen vegetables) => (pancakes), (frozen vegetables) => (burgers), (frozen vegetables) => (cake), (frozen vegetables) => (cookies), (frozen vegetables) => (escalope), (frozen vegetables) => (low fat yogurt), (frozen vegetables) => (shrimp), (frozen vegetables) => (tomatoes), (frozen vegetables) => (olive oil), (frozen vegetables) => (frozen smoothie), (frozen vegetables) => (turkey), (pancakes) => (mineral water), (pancakes) => (eggs), (pancakes) => (spaghetti), (pancakes) => (french fries), (pancakes) => (chocolate), (pancakes) => (green tea), (pancakes) => (milk), (pancakes) => (ground beef), (pancakes) => (frozen vegetables), (pancakes) => (cake), (pancakes) => (cookies), (pancakes) => (escalope), (pancakes) => (low fat yogurt), (pancakes) => (shrimp), (pancakes) => (tomatoes), (pancakes) => (olive oil), (pancakes) => (frozen smoothie), (pancakes) => (turkey), (burgers) => (mineral water), (burgers) => (eggs), (burgers) => (spaghetti), (burgers) => (french fries), (burgers) => (chocolate), (burgers) => (green tea), (burgers) => (milk), (burgers) => (ground beef), (burgers) => (frozen vegetables), (burgers) => (pancakes), (burgers) => (cake), (burgers) => (cookies), (burgers) => (escalope), (burgers) => (low fat yogurt), (burgers) => (shrimp), (burgers) => (tomatoes), (burgers) => (olive oil), (burgers) => (frozen smoothie), (burgers) => (turkey), (cake) => (mineral water), (cake) => (eggs), (cake) => (spaghetti), (cake) => (french fries), (cake) => (chocolate), (cake) => (green tea), (cake) => (milk), (cake) => (ground beef), (cake) => (frozen vegetables), (cake) => (pancakes), (cake) => (cookies), (cake) => (escalope), (cake) => (low fat yogurt), (cake) => (shrimp), (cake) => (tomatoes), (cake) => (olive oil), (cake) => (frozen smoothie), (cake) => (turkey), (cookies) => (mineral water), (cookies) => (eggs), (cookies) => (spaghetti), (cookies) => (french fries), (cookies) => (chocolate), (cookies) => (green tea), (cookies) => (milk), (cookies) => (ground beef), (cookies) => (frozen vegetables), (cookies) => (pancakes), (cookies) => (cake), (cookies) => (escalope), (cookies) => (low fat yogurt), (cookies) => (shrimp), (cookies) => (tomatoes), (cookies) => (olive oil), (cookies) => (frozen smoothie), (cookies) => (turkey), (escalope) => (mineral water), (escalope) => (eggs), (escalope) => (spaghetti), (escalope) => (french fries), (escalope) => (chocolate), (escalope) => (green tea), (escalope) => (milk), (escalope) => (ground beef), (escalope) => (frozen vegetables), (escalope) => (pancakes), (escalope) => (cake), (escalope) => (cookies), (escalope) => (low fat yogurt), (escalope) => (shrimp), (escalope) => (tomatoes), (escalope) => (olive oil), (escalope) => (frozen smoothie), (escalope) => (turkey), (low fat yogurt) => (mineral water), (low fat yogurt) => (eggs), (low fat yogurt) => (spaghetti), (low fat yogurt) => (french fries), (low fat yogurt) => (chocolate), (low fat yogurt) => (green tea), (low fat yogurt) => (milk), (low fat yogurt) => (ground beef), (low fat yogurt) => (frozen vegetables), (low fat yogurt) => (pancakes), (low fat yogurt) => (cake), (low fat yogurt) => (cookies), (low fat yogurt) => (escalope), (low fat yogurt) => (shrimp), (low fat yogurt) => (tomatoes), (low fat yogurt) => (olive oil), (low fat yogurt) => (frozen smoothie), (low fat yogurt) => (turkey), (shrimp) => (mineral water), (shrimp) => (eggs), (shrimp) => (spaghetti), (shrimp) => (french fries), (shrimp) => (chocolate), (shrimp) => (green tea), (shrimp) => (milk), (shrimp) => (ground beef), (shrimp) => (frozen vegetables), (shrimp) => (pancakes), (shrimp) => (cake), (shrimp) => (cookies), (shrimp) => (escalope), (shrimp) => (low fat yogurt), (shrimp) => (tomatoes), (shrimp) => (olive oil), (shrimp) => (frozen smoothie), (shrimp) => (turkey), (tomatoes) => (mineral water), (tomatoes) => (eggs), (tomatoes) => (spaghetti), (tomatoes) => (french fries), (tomatoes) => (chocolate), (tomatoes) => (green tea), (tomatoes) => (milk), (tomatoes) => (ground beef), (tomatoes) => (frozen vegetables), (tomatoes) => (pancakes), (tomatoes) => (cake), (tomatoes) => (cookies), (tomatoes) => (escalope), (tomatoes) => (low fat yogurt), (tomatoes) => (shrimp), (tomatoes) => (olive oil), (tomatoes) => (frozen smoothie), (tomatoes) => (turkey), (olive oil) => (mineral water), (olive oil) => (eggs), (olive oil) => (spaghetti), (olive oil) => (french fries), (olive oil) => (chocolate), (olive oil) => (green tea), (olive oil) => (milk), (olive oil) => (ground beef), (olive oil) => (frozen vegetables), (olive oil) => (pancakes), (olive oil) => (cake), (olive oil) => (cookies), (olive oil) => (escalope), (olive oil) => (low fat yogurt), (olive oil) => (shrimp), (olive oil) => (tomatoes), (olive oil) => (frozen smoothie), (olive oil) => (turkey), (frozen smoothie) => (mineral water), (frozen smoothie) => (eggs), (frozen smoothie) => (spaghetti), (frozen smoothie) => (french fries), (frozen smoothie) => (chocolate), (frozen smoothie) => (green tea), (frozen smoothie) => (milk), (frozen smoothie) => (ground beef), (frozen smoothie) => (frozen vegetables), (frozen smoothie) => (pancakes), (frozen smoothie) => (cake), (frozen smoothie) => (cookies), (frozen smoothie) => (escalope), (frozen smoothie) => (low fat yogurt), (frozen smoothie) => (shrimp), (frozen smoothie) => (tomatoes), (frozen smoothie) => (olive oil), (frozen smoothie) => (turkey), (turkey) => (mineral water), (turkey) => (eggs), (turkey) => (spaghetti), (turkey) => (french fries), (turkey) => (chocolate), (turkey) => (green tea), (turkey) => (milk), (turkey) => (ground beef), (turkey) => (frozen vegetables), (turkey) => (pancakes), (turkey) => (cake), (turkey) => (cookies), (turkey) => (escalope), (turkey) => (low fat yogurt), (turkey) => (shrimp), (turkey) => (tomatoes), (turkey) => (olive oil), (turkey) => (frozen smoothie)

Name:- Chetan Mandavkar
Roll No. T093
Subject:- Artificial Intelligence

Sheth L.U.J. And Sir M.V. College

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Association Rules:
...
133 (ground beef, eggs) (mineral water) 0.010132 0.506667
149 (milk, ground beef) (mineral water) 0.010865 0.503930
121 (ground beef, chocolate) (mineral water) 0.010932 0.473988
144 (milk, frozen vegetables) (mineral water) 0.011065 0.468927
100 (soup) (mineral water) 0.023064 0.456464
160 (pancakes, spaghetti) (mineral water) 0.011465 0.455026
158 (olive oil, spaghetti) (mineral water) 0.010265 0.447674
155 (milk, spaghetti) (mineral water) 0.015731 0.443609
124 (milk, chocolate) (mineral water) 0.013998 0.435685
152 (ground beef, spaghetti) (mineral water) 0.017064 0.435374
147 (spaghetti, frozen vegetables) (mineral water) 0.011998 0.430622
135 (milk, eggs) (mineral water) 0.013065 0.424242
95 (olive oil) (mineral water) 0.027596 0.419028
153 (ground beef, mineral water) (spaghetti) 0.017064 0.416938
77 (ground beef) (mineral water) 0.040928 0.416554
115 (eggs, chocolate) (mineral water) 0.013465 0.405622
130 (spaghetti, chocolate) (mineral water) 0.015865 0.404762
98 (salmon) (mineral water) 0.017064 0.401254
11 (cereals) (mineral water) 0.010265 0.398964
79 (ground beef) (spaghetti) 0.039195 0.398915

lift
133 2.125563
149 2.110308
121 1.988472
144 1.967236
100 1.914955
160 1.908923
158 1.878079
155 1.861024
```

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147 (spaghetti, frozen vegetables) (mineral water) 0.011998 0.430622
135 (milk, eggs) (mineral water) 0.013065 0.424242
95 (olive oil) (mineral water) 0.027596 0.419028
153 (ground beef, mineral water) (spaghetti) 0.017064 0.416938
77 (ground beef) (mineral water) 0.040928 0.416554
115 (eggs, chocolate) (mineral water) 0.013465 0.405622
130 (spaghetti, chocolate) (mineral water) 0.015865 0.404762
98 (salmon) (mineral water) 0.017064 0.401254
11 (cereals) (mineral water) 0.010265 0.398964
79 (ground beef) (spaghetti) 0.039195 0.398915

lift
133 2.125563
149 2.110308
121 1.988472
144 1.967236
100 1.914955
160 1.908923
158 1.878079
155 1.861024
124 1.827780
152 1.826477
147 1.806541
135 1.779778
95 1.757904
153 2.394681
77 1.747522
115 1.701663
130 1.698053
98 1.683336
11 1.673729
79 2.291162
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